

AJAY VENKATESH

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Education

New York University | Master of Science in Computer Engineering (GPA – 3.9/4)

Sep 2024 – May 2026

• Coursework: System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

Skills

Programming Languages: C++, Java, Python, TypeScript, JavaScript (ES6), SQL

Web Technologies: HTML, CSS, React, REST API, GraphQL API, Microservices, Node.js, Express.js, Sequelize, Spring, Spring Boot

Database: RDBMS (MySQL, PostgreSQL), NoSQL (Dynamo DB, FireBase, MongoDB), Dataverse

Cloud: AWS (AWS CDK, EC2, S3, Dynamo DB, API Gateway, Lambda, Cognito, IAM, Q, SageMaker), Azure (OpenAI services, Function Apps, Security Data lakes, Storage blobs), GRPC, Jenkins

Tools: Git, Visual Studio, Cursor, Jupyter, Postman, Docker, Kubernetes, Power BI, QuickSight, JUnit

AI/ML: LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning, RL.

Experience

CampusMesh

Jan 2026 – Present

Software Engineer | Real-time microservices, recommendation engine, and LLM-based AI retrieval pipelines

- Developed scalable **Node.js microservices** powering real-time academic workflows across study groups, messaging, calls, notifications, and profile management, backed by **MySQL (Sequelize)** and **DynamoDB**.
- Deployed **serverless architecture** on AWS using **Serverless Framework** with **Lambda, API Gateway, S3, and CloudFront** across **4 environments** (local, dev, staging, production); enforced security with **KMS-managed keys, SSM Parameter Store**, and **least-privilege IAM**.
- Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.

Amazon

May 2025 – Aug 2025

Software Development Engineer Intern | Serverless AWS infrastructure, React 18 frontend, and Java backend systems

- Architected a secure UI console for Amazon fulfillment center teams to view image/video evidence across FCs; presented the **system design** for approval and established Role Based Access (**4+ roles**) using **OAuth, Midway SSO, and HELIS**.
- Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, DynamoDB, S3, and IAM, reducing provisioning time by **70%** and enabling **multi-stage, multi-region CI/CD deployments**.
- Built a **React 18** frontend with optimized API integration, advanced filtering, client-side caching, reducing page load time by **40%**.
- Engineered backend services in **Java** using **Coral, Smithy, Dagger**, and **CBOR serialization**; applied low-level design and design patterns, reduced backend execution time by **35%** and ensured reliability by unit and integration testing using **Mockito**.

New York University (Novel AI Technologies, Inc.)

Aug 2024 – Present

Software Developer (On-Campus) | NSF-funded AI health platform, ETL pipelines, and full-stack insurance analytics

- Architected an **NSF-funded (\$100K)** AI-powered health app for **iOS and Android**, processing real-time wearable data; deployed cloud-hosted AI models for health metrics analysis and food image processing.
- Engineered ETL pipelines transforming **NoSQL** into **PostgreSQL** using **concurrency control and locking mechanisms**; processed **250K+** records and powered **AWS QuickSight** dashboards for real-time health analytics.
- Developed an insurance analytics platform (**React, Node.js, REST APIs**); deployed **Docker** containers on **EC2**, implemented Stripe billing, RBAC, and row-level security, generating **\$50K+** revenue.
- Streamlined underwriting via broker portal integrated with insurer platform; implemented **document and media processing** and **WebSocket-based real-time communication**; secured **6 insurance clients**.

PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Senior Software Engineer | Cloud-native backend, NLP/OCR automation, and data engineering on Azure

- Developed a distributed backend on **Microsoft Azure** for a health platform, integrating with **Microsoft CRM** and implementing event-driven processing and fault tolerance.
- Led code reviews and mentored engineers on **software design principles**, reducing production bugs by **25%**.
- Deployed an **NLP based OCR neural model** on Azure to automate structured data extraction, reducing data processing time by **75%**.
- Optimized data ingestion with **parallel processing** for **100K-row** from Excel; reduced execution time from **5 hours to 1 hour**.
- Built **Power BI dashboards** processing large datasets from multiple databases, improving reporting efficiency by **40%**.

Projects

Advanced Expense Tracker (Java Desktop Application)  | Java, JavaFX, JDBC, SQLite, JFreeChart

- Architected a modular **Java application** supporting individual and group expenses with secure login, applying **OOP principles** and **MVC & DAO design patterns**.
- Implemented concurrent transaction handling and real-time cost splitting using **multithreading** and synchronized blocks to ensure thread-safe database updates.

Hospital Management System (Full-Stack Application) | React, Node.js, Express, MongoDB, SageMaker, Hugging Face

- Built a **full-stack healthcare platform** using **React, Node.js, Express, and MongoDB** to manage appointments, patient records, lab reports, and medical history with **role-based access control**.
- Fine-tuned a **Hugging Face transformer-based model** achieving **88% accuracy** for health condition prediction and deployed it on **AWS SageMaker** for real-time diagnostic support in the doctor dashboard.

Certifications

AWS Cloud Practitioner 

Power BI Data Analyst Associate 

Azure Data Fundamentals 

Microsoft Developer Associate 

Microsoft Solution Architect Expert 

Stanford Machine Learning (Coursera) 